20. (a) Using the mirror equation and the formula of magnification, deduce that "the virtual image produced by a convex mirror is always diminished in size and is located between the pole and the focus." CBSE 2025 *2*

20. (a) Using the mirror equation and the formula of magnification, deduce that "the virtual image produced by a convex mirror is always diminished in size and is located between the pole and the focus." CBSE 2025 *2*

Let, $u = nf$	
From the Mirror formula,	
$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$	½
$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$	
$=\frac{1}{f}+\frac{1}{nf}$	
$v=\frac{nf}{n+1}$	
$n=+ve \quad \therefore v<f$	½
$m=\frac{-v}{u}=\frac{1}{n+1}$	½
m is always positive & less than 1.	½
Note: **Award full credit if correctly concluded by any other method.**	

25. (a) A concave mirror has radius of curvature 20 cm. Calculate the distance of an object from the mirror so as to form an image of magnification – 2. Also find the location of the image.

(b) If the silver coating around the centre of a concave mirror is removed, will the mirror still form the image of an object ? Justify your answer. CBSE 2025 *3*

25. (a) A concave mirror has radius of curvature 20 cm. Calculate the distance of an object from the mirror so as to form an image of magnification – 2. Also find the location of the image.

(b) If the silver coating around the centre of a concave mirror is removed, will the mirror still form the image of an object ? Justify your answer. CBSE 2025 *3*

(a) $m = -\frac{v}{u}$	½
$-2 = -\frac{v}{u}$	
$v = 2u$	
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	
$\frac{1}{-10} = \frac{1}{2u} + \frac{1}{u}$	½
$u = -15\,\text{cm}$	½
$v = -30\,\text{cm}$	½
(b) Yes, same image is formed with reduced intensity, because reflecting area get reduced and laws of reflection still hold good for remaining part of the mirror.	1

(ii) Use mirror formula to deduce that a convex mirror always produces a virtual image of an object kept in front of it.

(ii) Use mirror formula to deduce that a convex mirror always produces a virtual image of an object kept in front of it.

(ii)	$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$	½
	$u<0$, $f>0$	
	$\frac{1}{v}+\frac{1}{(-u)}=\frac{1}{f}$	½
	$\frac{1}{v}=\frac{1}{f}+\frac{1}{u}$	½
$\frac{1}{v}$ is positive		
$\therefore$ v is positive $\Rightarrow$ virtual image		½

26. Draw a ray diagram showing the image formation when a concave mirror produces a real, inverted and magnified image of an object and hence obtain the mirror formula. CBSE 2025 *3*

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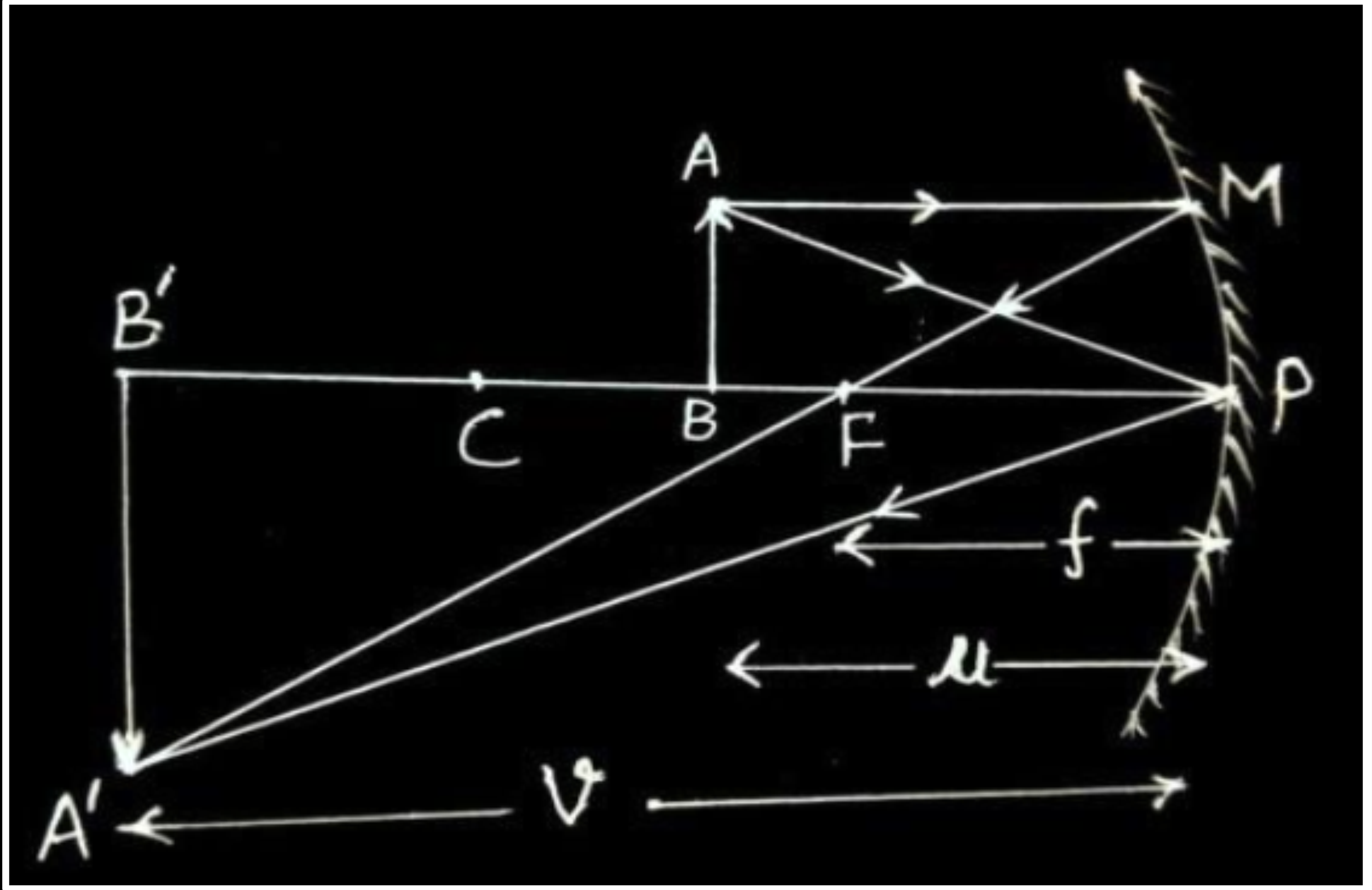


Note: Please deduct ½ mark of this diagram if not showing arrows with the rays.

In similar triangles

$\Delta A'B'F$ and ΔMPF 1

$$\frac{A'B'}{MP}=\frac{B'F}{FP}$$

or $\dfrac{A'B'}{AB}=\dfrac{B'F}{FP}$ $(\because MP=AB)$ ---------(1) ½

In similar triangles $\Delta A'B'P$ and ΔABP

$\dfrac{A'B'}{AB}=\dfrac{PB'}{PB}$ -------(2) ½

from equation (1) and (2)

$$\frac{B'F}{FP}=\frac{PB'}{PB}$$

$$\frac{PB'-PF}{FP}=\frac{PB'}{PB}$$

$\dfrac{(-v)-(-f)}{(-f)}=\dfrac{(-v)}{(-u)}$ ½

on solving we get

$\dfrac{1}{v}+\dfrac{1}{u}=\dfrac{1}{f}$ ½

18. (a) The real image of an object is formed at a distance twice the distance of the object from a concave mirror of focal length 10 cm. Find the distance between the object and the image. **2**

CBSE 2025

18. (a) The real image of an object is formed at a distance twice the distance of the object from a concave mirror of focal length 10 cm. Find the distance between the object and the image. **2**

CBSE 2025

$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$	½
$u=-x,\ v=-2x$	½
$\frac{1}{-10}=\frac{1}{-x}+\frac{1}{-2x}=-\frac{3}{2x}$	½
$x=15\,\text{cm}$	
Distance between object and image, d = 2x- x = 30- 15 = 15cm	½

9. The magnification produced by a spherical mirror is $-2 \cdot 0$. The mirror used and the nature of the image formed will be

(A) Convex and virtual

(B) Concave and real

(C) Concave and virtual

(D) Convex and real

CBSE 2025

9. The magnification produced by a spherical mirror is – 2·0. The mirror used and the nature of the image formed will be

(A) Convex and virtual

(B) Concave and real

(C) Concave and virtual

(D) Convex and real

CBSE 2025

(B) Concave and real

32. (a) (i) A thin pencil of length (f/4) is placed coinciding with the principal axis of a mirror of focal length f. The image of the pencil is real and enlarged, just touches the pencil. Calculate the magnification produced by the mirror.

CBSE 2025

(i) As the pencil lies between f and 2f such that one end of the pencil coincides with 2f.	
Position of the other end $(u) = -\left(2f - \frac{f}{4}\right) = -\frac{7f}{4}$	½
Magnification $(m) = \frac{f}{f - u}$	½
$= \frac{-f}{-f - \left(-\frac{7f}{4}\right)}$	½
$m = -\frac{4}{3}$	1

Alternatively: -

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{4}{7f} = -\frac{1}{f}$$

$$\frac{1}{v} = -\frac{1}{f} + \frac{4}{7f}$$

$$v = -\frac{7f}{3}$$

$$m = -\frac{v}{u} = -\frac{4}{3}$$

9. The focal length of a concave mirror in air is f. When the mirror is immersed in a liquid of refractive index $\frac{5}{3}$, its focal length will become **1**

(A) $\frac{5}{3}f$

(B) $\frac{3}{5}f$

(C) $\frac{2}{3}f$

(D) f

CBSE 2025

9. The focal length of a concave mirror in air is f. When the mirror is immersed in a liquid of refractive index $\frac{5}{3}$, its focal length will become **1**

(A) $\frac{5}{3}$ f

(B) $\frac{3}{5}$ f

(C) $\frac{2}{3}$ f

(D) f

CBSE 2025

(D) f

(ii) An object is placed between the pole and the focus of a concave mirror. Using mirror formula, prove mathematically that it produces a virtual and an enlarged image. CBSE 2024

(ii) An object is placed between the pole and the focus of a concave mirror. Using mirror formula, prove mathematically that it produces a virtual and an enlarged image.

CBSE 2024

(ii) $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$ $v=\frac{uf}{u-f}$ Following new cartesian sign conversion $v=\frac{(-u)(-f)}{-u-(-f)}$ $v=\frac{uf}{f-u}$ as $f > u$	1
v is +ve, So image is virtual. $m=-\frac{v}{u}=\frac{f}{f-u}>1$ i.e. Enlarged image	1

19. The radius of curvature of a convex mirror is 30 cm. It forms an image of an object which is half the size of the object. Find the separation between the object and the image.

CBSE 2024

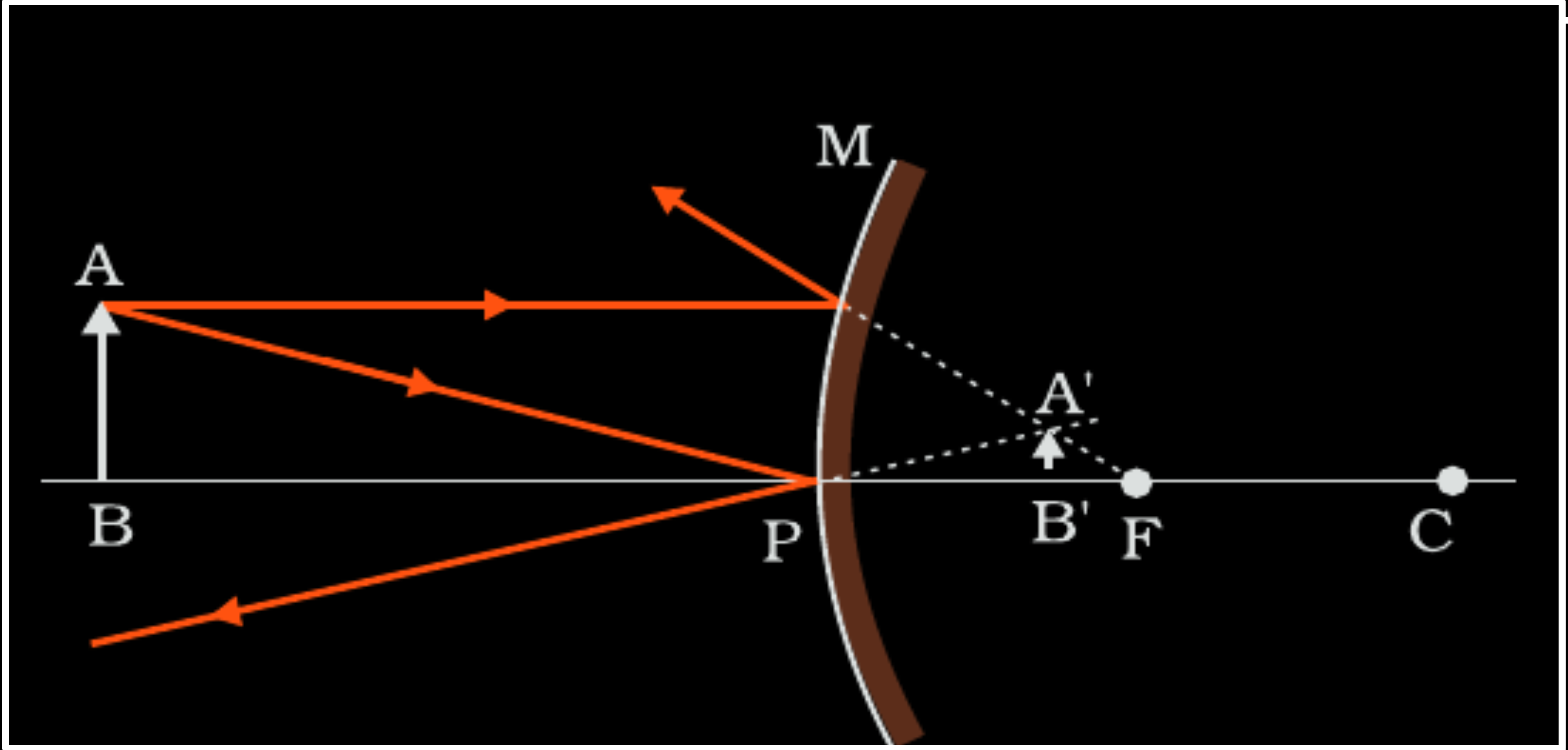
19. The radius of curvature of a convex mirror is 30 cm. It forms an image of an object which is half the size of the object. Find the separation between the object and the image.

CBSE 2024

$m = -\frac{v}{u} = \frac{h_I}{h_O} = \frac{1}{2}$	½
$u = -2v$	
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	½
$\frac{1}{15} = \frac{1}{v} - \frac{1}{2v}$	
On solving	
$\lvert v \rvert = 7.5\,\text{cm}$	
$\lvert u \rvert = +15.0\,\text{cm}$	
Separation = 15.0 + 7.5	½
= 22.5 cm	½

32. (a) (i) Draw a ray diagram for the formation of the image of an object by a convex mirror. Hence, obtain the mirror equation.

CBSE 2024



For paraxial rays MP can be considered to be a straight line perpendicular to CP,
Therefore right angled triangles $A'B'F$ and MPF are similar 1

$$\frac{B'A'}{PM} = \frac{B'F}{FP}$$

Or $\frac{B'A'}{BA} = \frac{B'F}{FP}$ ($\because$ PM = AB) ----------------(1) ½

Since $\angle$ APB = $\angle$ $A'PB'$, the right angled triangles $A'PB'$ and ABP are also similar

Therefore , $\frac{B'A'}{BA} = \frac{B'P}{BP}$ ----------------------------------- (2) ½

Comparing eq (1) and (2), we get

$$\frac{B'F}{FP} = \frac{B'P}{BP}$$

$$\frac{PF - PB'}{FP} = \frac{B'P}{BP}$$

Using sign convention

PF = f, PB'= +v , PB = -u ½

on solving $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ ½

19. Explain the following : *2*

(b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ? CBSE 2024

19. Explain the following : *2*

(b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ? CBSE 2024

(b) Yes,	½
Rays converging to a point behind a plane or convex mirror are reflected to a point in front of the mirror on a screen	½

14. *Assertion (A) :* Plane and convex mirrors cannot produce real images under any circumstance.

Reason (R) : A virtual image cannot serve as an object to produce a real image.

CBSE 2024

14. *Assertion (A) :* Plane and convex mirrors cannot produce real images under any circumstance.

Reason (R) : A virtual image cannot serve as an object to produce a real image.

CBSE 2024

(D) Both Assertion (A) and Reason (R) are false.

18. An object is placed 30 cm in front of a concave mirror of radius of curvature 40 cm. Find the (i) position of the image formed and (ii) magnification of the image. CBSE 2024 *2*

18. An object is placed 30 cm in front of a concave mirror of radius of curvature 40 cm. Find the (i) position of the image formed and (ii) magnification of the image. CBSE 2024 *2*

(i) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	½
$\frac{1}{v} + \frac{1}{-30} = \frac{1}{-20}$	
On solving	½
v = - 60 cm	
(ii) m = - $\frac{v}{u}$	½
$= -\left(\frac{-60}{-30}\right) = -2$	½

(ii) An object is kept 20 cm from a convex mirror of radius of curvature 60 cm. Find the position of the image formed. Will the image be real or virtual ?

CBSE 2024

(ii) An object is kept 20 cm from a convex mirror of radius of curvature 60 cm. Find the position of the image formed. Will the image be real or virtual ?

CBSE 2024

(ii) u= -20cm	
$f = \frac{R}{2} = 30\text{cm}$	
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	½
$\frac{1}{v} + \left(-\frac{1}{20}\right) = \frac{1}{30}$	½
$\frac{1}{v} = \frac{1}{30} + \frac{1}{20} = \frac{3+2}{6} = \frac{5}{60}$	
$v = 12\,\text{cm}$	½
Virtual Image.	½

33. (a) (i) Draw a ray diagram to show the formation of image by a concave mirror when the object is placed between f and 2f. Using this diagram, derive the mirror equation.

(ii) An object is kept 30 cm in front of a convex mirror of focal length 12 cm. Find the nature and position of the image formed. *5*

CBSE 2023 C

From $\triangle A'B'C' \sim \triangle DPF$

$$\frac{B'A'}{BA} = \frac{B'F}{FP} (\because PD = AB) \quad \text{---------(1)}$$

½

Form $\triangle A'B'P \sim \triangle ABP$

$$\frac{B'A'}{BA} = \frac{B'P}{BP} \quad \text{---------(2)}$$

½

Comparing equation (1) and (2)

$$\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{BP}$$

Here B’P = - v

FP = -f

BP = - u

$$\frac{-v+f}{-f} = \frac{-v}{-u}$$

½

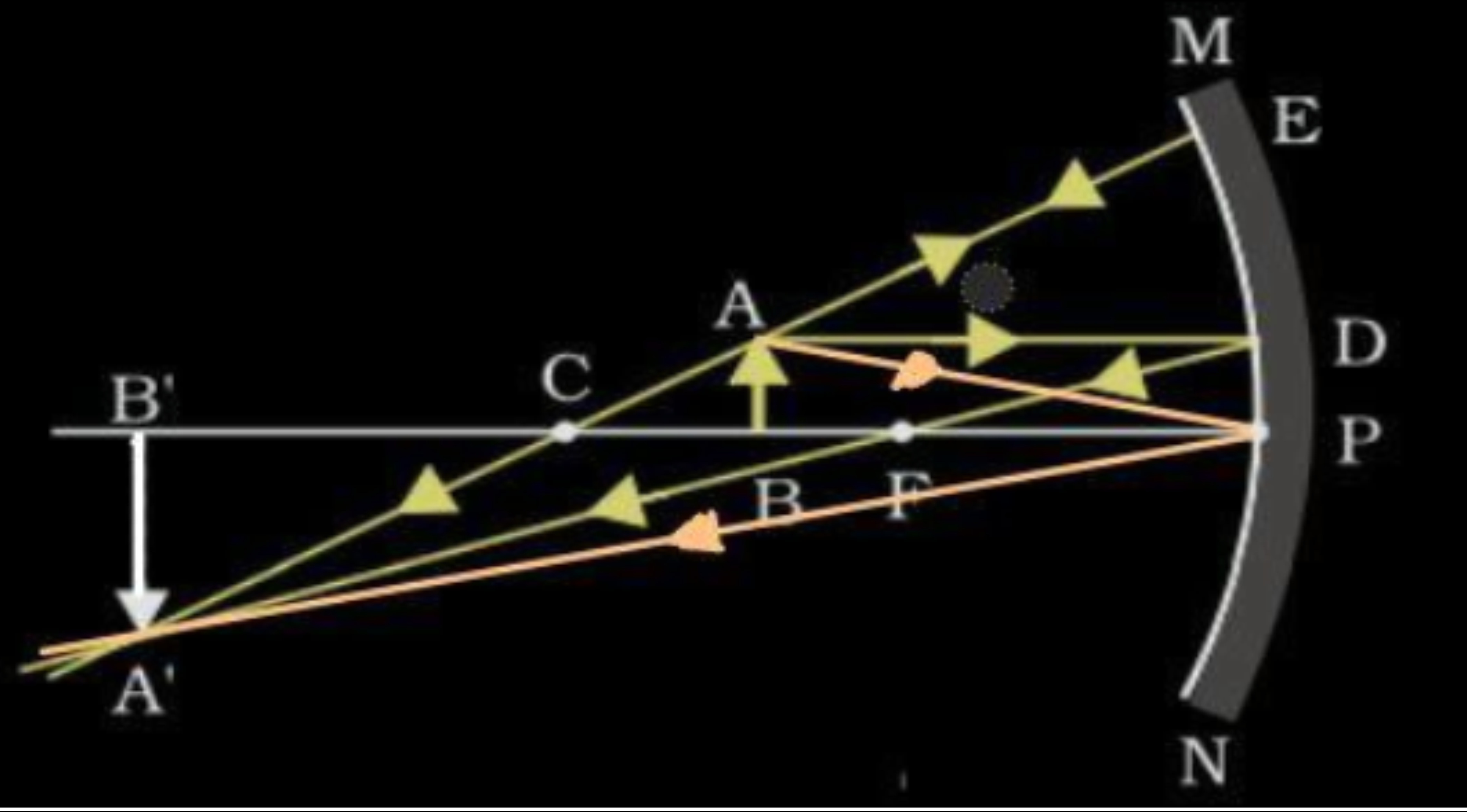
$$\frac{v-f}{f} = \frac{v}{u}$$

$$\frac{v}{f} = 1 + \frac{v}{u}$$

On solving we get

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

½



(ii) u= -30 cm

f= + 12 cm

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

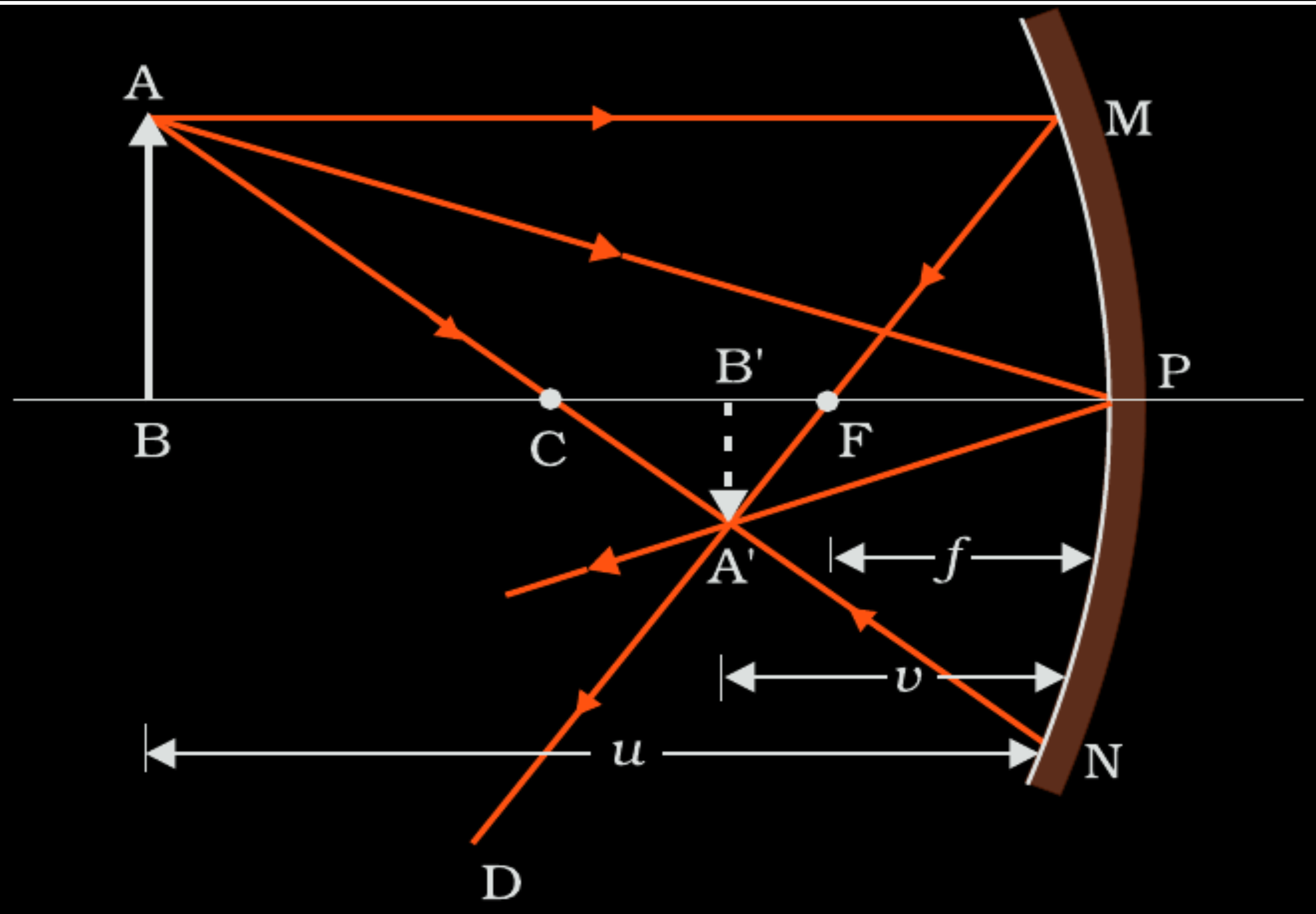
$$\frac{1}{v} = \frac{1}{12} + \frac{1}{30} = \frac{7}{60}$$ 1

v= $\frac{60}{7}$ cm ½

Since v is positive image is virtual and erect.

31. (a) Draw a ray diagram for formation of a real and diminished image of an object kept in front of a concave mirror. Hence derive the mirror equation.

(b) A concave mirror of focal length 10 cm produces a real image which is 3 times the size of the object. Find the distance of the object from the mirror. CBSE 2023 C *5*



1

$$\Delta A'B'F \sim \Delta MPF$$

$$\frac{B'A'}{BA}=\frac{B'F}{FP} \quad (\because PM = AB) \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ (1)$$

½

Also, $\Delta A'B'P \sim \Delta ABP$

$$\frac{B'A'}{BA}=\frac{B'P}{BP} \quad \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ (2)$$

½

Comparing eq. (1) and (2)

$$\frac{B'F}{FP}=\frac{B'P-FP}{FP}=\frac{B'P}{FP}$$

½

$$B'P=-v$$

$$FP=-f$$

$$BP=-u$$

On solving we get , $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$ ½

(b)

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$$ ½

$$\frac{1}{u}=\frac{1}{f}-\frac{1}{v}$$ ½

$$-\frac{v}{u}=-3$$

$$v=3u$$ ½

$$\frac{1}{u}=\frac{1}{f}-\frac{1}{3u}$$

$$\frac{1}{u}+\frac{1}{3u}=\frac{1}{f}$$ ½

$$\frac{4}{3u}=\frac{1}{f}$$ ½

$$u=\frac{4f}{3}=-\frac{40}{3}cm$$

10. An object is placed at a distance of 10 cm in front of a concave mirror of radius of curvature 15 cm. The nature and position the image formed is :

(a) real and inverted, 30 cm in front of the mirror

(b) real and inverted, 15 cm in front of the mirror

(c) virtual and erect, 30 cm behind the mirror

(d) virtual and erect, 15 cm behind the mirror

CBSE 2023

10. An object is placed at a distance of 10 cm in front of a concave mirror of radius of curvature 15 cm. The nature and position the image formed is :

(a) real and inverted, 30 cm in front of the mirror

(b) real and inverted, 15 cm in front of the mirror

(c) virtual and erect, 30 cm behind the mirror

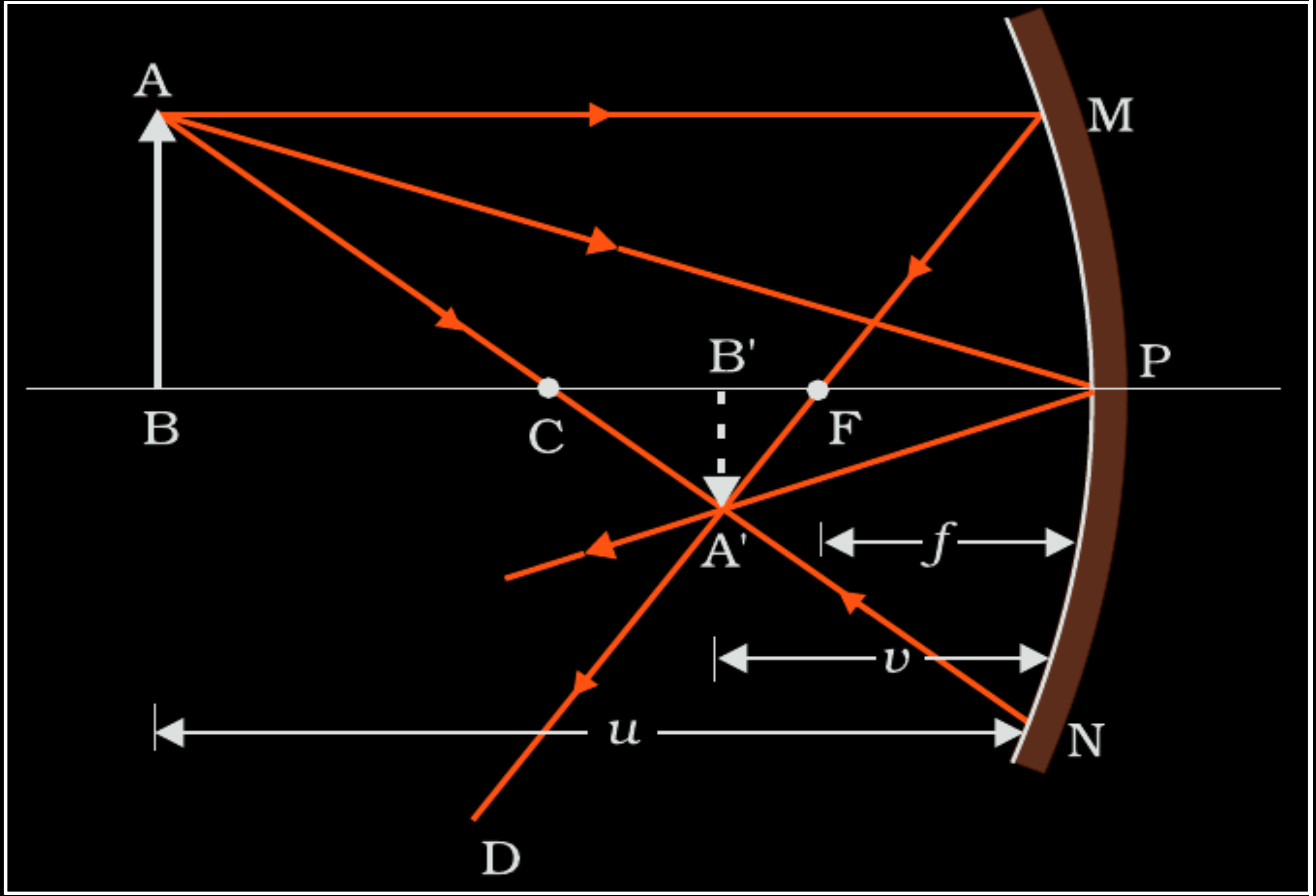
(d) virtual and erect, 15 cm behind the mirror

CBSE 2023

(a) real and inverted, 30 cm in front of the mirror

31. (a) (i) Draw a ray diagram showing the formation of a real image of an object placed at a distance 'u' in front of a concave mirror of radius of curvature 'R'. Hence, obtain the relation for the image distance 'v' in terms of u and R.

CBSE 2023



From Fig. the two right-angled triangles A′B′F and MPF are similar. (For paraxial rays, MP can be considered to be a straight line perpendicular to CP.) Therefore,	
$\frac{B'A'}{PM} = \frac{B'F}{FP}$ or $\frac{B'A'}{BA} = \frac{B'F}{FP}$ $(\because PM = AB)$ -----(i)	½
Since ∠ APB = ∠ A′PB′, the right angled triangles A′B′P and ABP are also similar. Therefore, $\frac{B'A'}{BA} = \frac{B'P}{BP}$ ----(ii)	½
Comparing equations (i) and (ii) $\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{BP}$ -----(iii)	½
B′ P = – v, FP = – f, BP = – u;	
Using these in Eq.(iii) we get $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$	½
Alternatively:- If the result derived by any other method, full credit to be given.	

9. For a concave mirror of focal length 'f', the minimum distance between the object and its real image is :

(a) zero

(b) f

(c) 2f

(d) 4f

CBSE 2023

9. For a concave mirror of focal length 'f', the minimum distance between the object and its real image is :

(a) zero   (b) f

(c) 2f   (d) 4f

CBSE 2023

(a) Zero

(ii) A concave mirror of focal length 12 cm forms a three times magnified virtual image of an object. Find the distance of the object from the mirror.

CBSE 2023

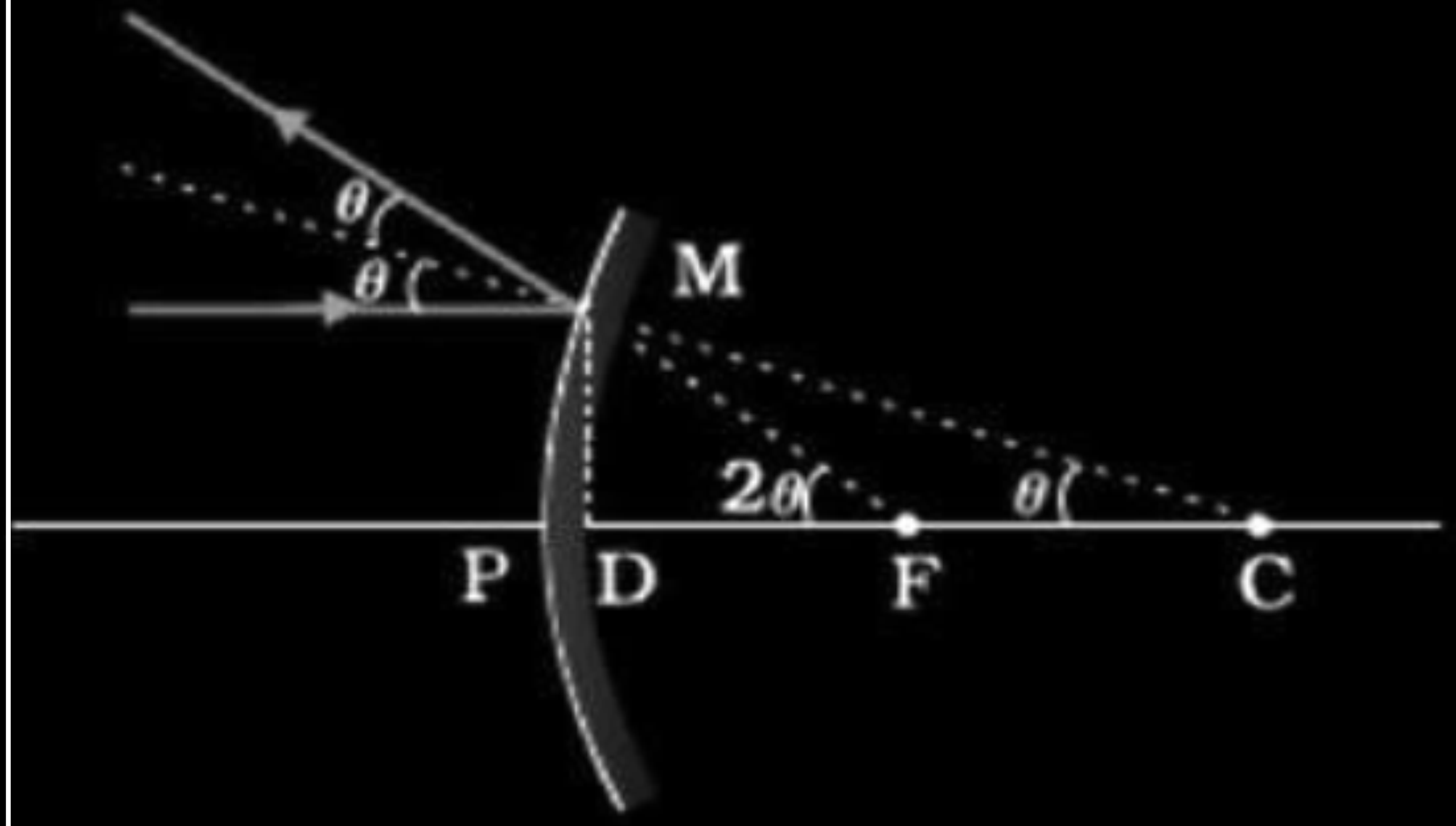
(ii) A concave mirror of focal length 12 cm forms a three times magnified virtual image of an object. Find the distance of the object from the mirror.

CBSE 2023

(ii) $m = +3$, f = -12 cm , $u = ?$	
$m = -\frac{v}{u} = 3 \Rightarrow v = -3u$	½
using mirror formula	
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	½
$\frac{1}{-3u} + \frac{1}{u} = \frac{1}{-12}$	½
	½
u = - 8 cm	

35. (a) Define the term 'focal length of a mirror'. With the help of a ray diagram, obtain the relation between its focal length and radius of curvature.

CBSE 2020

	(a) **Focal length of mirror**: It is the distance of the point from the pole of mirror through which ray of light moving parallel to its principle axis passes (or appear to come from). **Alternatively:** It is half of the distance of its centre of curvature from the pole of a mirror.	1
 	Let C be the centre of curvature of mirror, MD be the perpendicular from M to the principal axis. $\angle MCP = \theta \text{ and } \angle MFP = 2\theta$ $\tan\theta = \frac{MD}{CD}$, $\tan 2\theta = \frac{MD}{FD}$ (1) For small angles, $\tan\theta \approx \theta$ and $\tan 2\theta \approx 2\theta$ From equation 1, $\frac{MD}{FD} = 2\frac{MD}{CD}$ $FD = \frac{CD}{2}$ ----equation (2) For small θ, the point D is very close to the point P $\therefore$ FD $\simeq$ FP $= f$ and CD $\simeq$ CP $= R$ $\therefore$ from equation 2, we get $f = \frac{R}{2}$	½ ½ ½

34. Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm, form real images of equal size. Size of object P is three times size of object Q. If the distance of P is 50 cm from the mirror, find the distance of Q from the mirror. CBSE 2020 **3**

34. Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm, form real images of equal size. Size of object P is three times size of object Q. If the distance of P is 50 cm from the mirror, find the distance of Q from the mirror. CBSE 2020 **3**

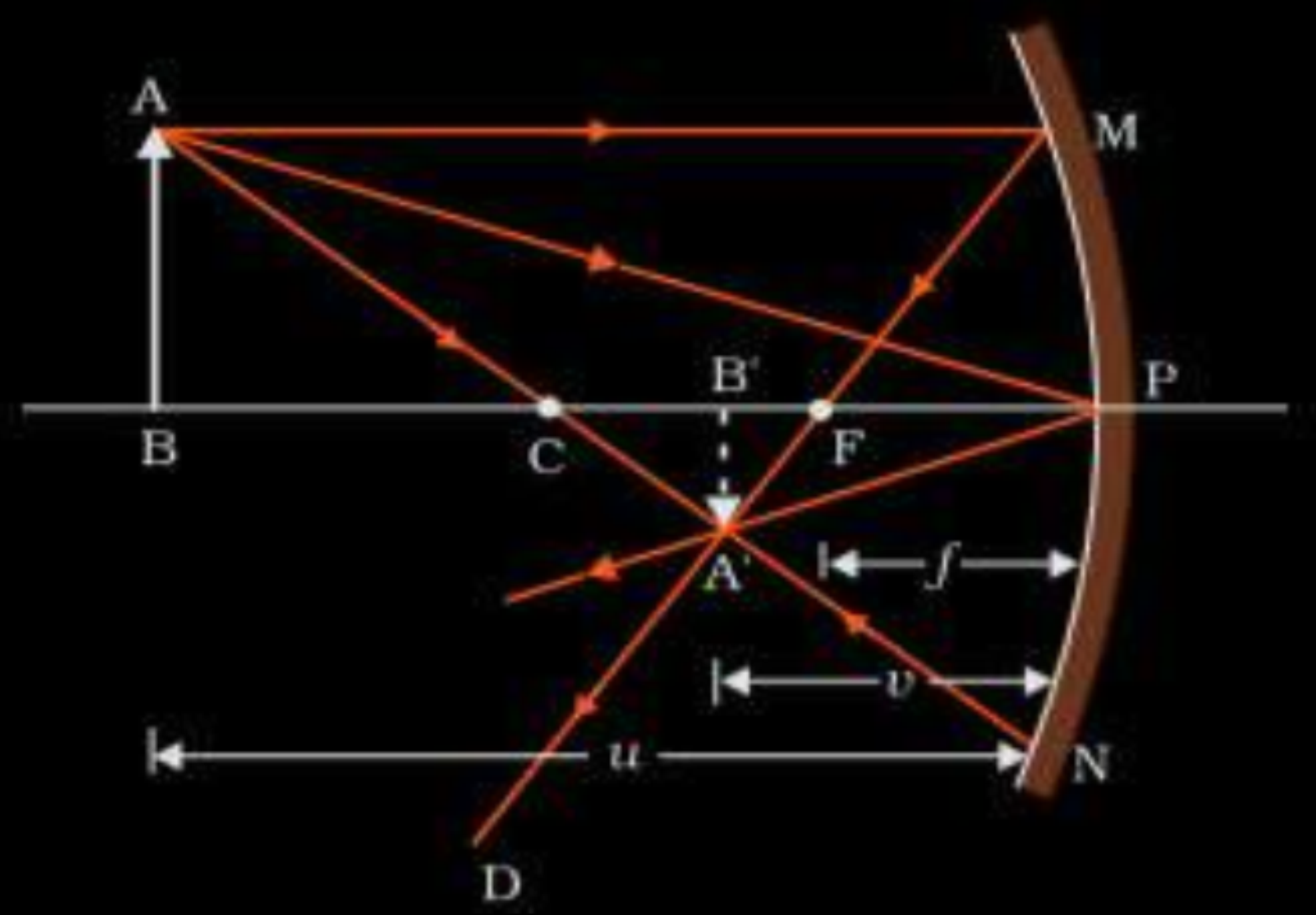
For object P $$m = \frac{h_2}{h_1} = \frac{f}{f - u_1}$$	½
For Object Q $$m' = \frac{h_2'}{h_1'} = \frac{f}{f - u_2}$$	½
Now $h_{1=}3h_1'$; $h_2 = h_2'$; $f = -20\ cm$; $u_1 = -50\ cm$ $$\frac{m}{m'} = \frac{h_2}{h_1} \times \frac{h_1'}{h_2'} = \frac{f - u_2}{f - u_1}$$	½
$$\frac{m}{m'} = \frac{1}{3} = \frac{-20 - u_2}{-20 + 50}$$	1
$$10 = -20 - u_2 \Rightarrow u_2 = -30\ cm$$	½

30. (a) Using the necessary ray diagram, derive the mirror formula for a concave mirror.

(b) In the magnified image of a measuring scale (with equidistant markings) lying along the principal axis of a concave mirror, the markings are not equidistant. Explain. *3*

CBSE 2020

a) Ray diagrams for concave mirror



½

Derivation of Mirror Formula

From the diagram,

$\Delta A'B'F$ & ΔMPF are similar

$$\therefore \frac{B'A'}{PM} = \frac{B'F}{FP}$$

$$\frac{B'A'}{BA} = \frac{B'F}{FP} \; (\because PM = AB) - - - - - - - - eq1$$

½

Since

$$\angle APB = \angle A'PB'$$

$\Delta A'B'P$ & ΔABP are also similar

$$\frac{B'A'}{BA} = \frac{B'P}{BP} \quad - - - - - - - - - - eq\ 2$$

½

Comparing eq. 1 and eq. 2

$$\frac{B'P}{BP} = \frac{B'P - FP}{FP}$$

As per the sign convention

$B'P = -v,\quad FP = -f,\quad$ BP = -u

$$\frac{-v+f}{-f} = \frac{-v}{-u} = \frac{v}{u}$$

$$-vu + uf = -vf$$

Dividing by uvf

$$\Rightarrow \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

1

b) Magnification is different for different object distances

½

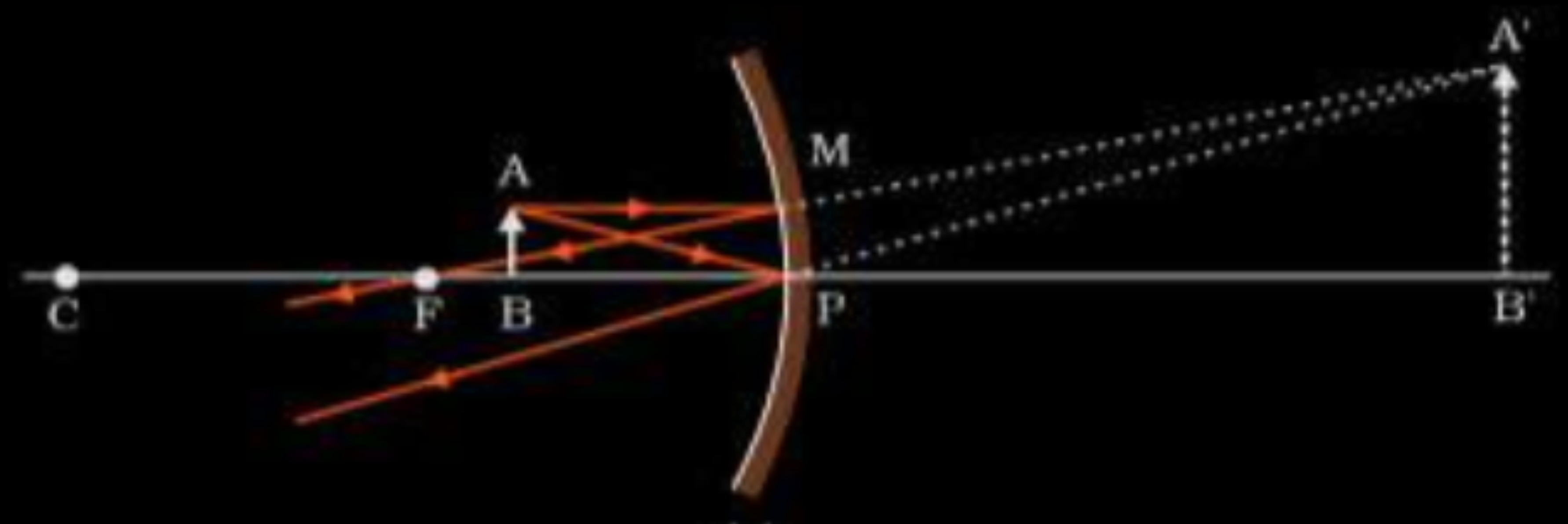
25. An object is kept 20 cm in front of a concave mirror of radius of curvature 60 cm. Find the nature and position of the image formed. CBSE 2020 *2*

25. An object is kept 20 cm in front of a concave mirror of radius of curvature 60 cm. Find the nature and position of the image formed. CBSE 2020 *2*

The focal length $f = \frac{-R}{2} = -30\ cm\ u = -20\ cm$	½
$$\frac{1}{V} + \frac{1}{u} = \frac{1}{f}$$	½
$$\therefore \frac{1}{V} - \frac{1}{20} = -\frac{1}{30} \quad \Rightarrow \frac{1}{V} = -\frac{1}{30} + \frac{1}{20}$$ $$\therefore V = +60\ cm$$	½
Nature of image: virtual, erect and magnified	½

(a) An object is placed in front of a concave mirror. It is observed that a virtual image is formed. Draw the ray diagram to show the image formation and hence derive the mirror equation $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$.

CBSE 2020



(a)

1

$$\Delta ABP \sim A'B'P$$

$$\frac{A'B'}{AB} = \frac{PB'}{PB} - - - - - - - - - - 1$$ ½

Also $\Delta A'B'F \sim MNP$ (for small curvature)

$$\therefore \frac{A'B'}{MP} = \frac{B'F}{PF}$$

$$\frac{A'B'}{AB} = \frac{B'F}{PF} - - - - - - - - - - - 2$$

From 1 and 2

$$\frac{PB'}{PB} = \frac{B'F}{PF} - - - - - - - - - - - - 3$$ ½

$$\frac{PB'}{PB} = \frac{B'P + PF}{PF} - - - - - - - - 4$$ ½

$$PB = -u \qquad PB' = v \qquad PF = -f$$

$$\frac{v}{-u} = \frac{v-f}{-f}$$

$$-vf = -vu + uf$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$ ½

An object is kept in front of a concave mirror of focal length 15 cm. The image formed is real and three times the size of the object. Calculate the distance of the object from the mirror.

CBSE 2019

An object is kept in front of a concave mirror of focal length 15 cm. The image formed is real and three times the size of the object. Calculate the distance of the object from the mirror.

CBSE 2019

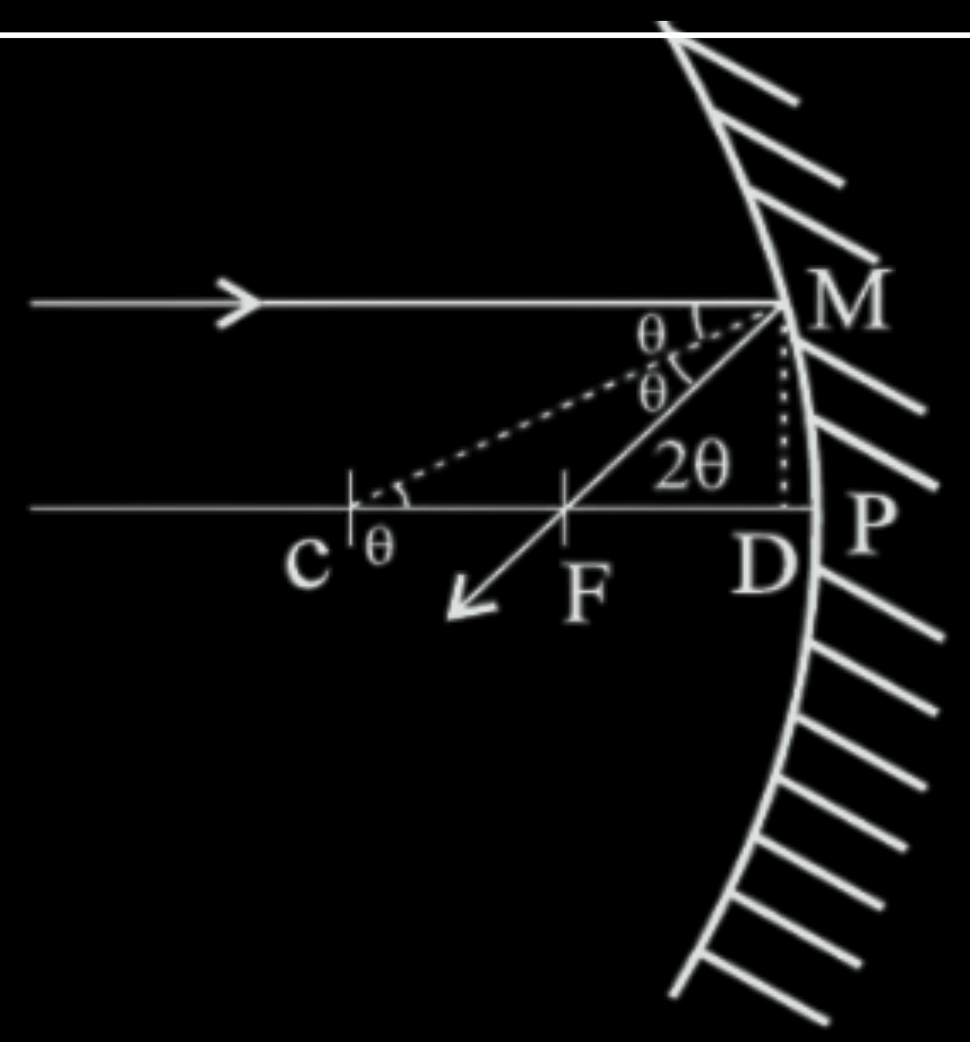
Here, $m = +3$ and $f = -15$ cm	
$m = \frac{v}{u} = 3 \quad \therefore \quad v = 3u$	½
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	½
$\frac{1}{-15} = \frac{1}{3u} + \frac{1}{u}$	½
$\Rightarrow \quad u = -20$ cm	½

For paraxial rays, show that the focal length of a spherical mirror is one-half of its radius of curvature.

CBSE 2019

For paraxial rays, show that the focal length of a spherical mirror is one-half of its radius of curvature.

CBSE 2019



1

From ray diagram

$$\tan\theta = \frac{MD}{CD} \quad \text{and } \tan 2\theta = \frac{MD}{FD}$$ ½

for small $\theta, \tan\theta \cong \theta$, $\tan 2\theta \cong 2\theta$

$$\therefore \frac{MD}{FD} = 2\frac{MD}{CD}$$

$$FD = \frac{CD}{2} \cong \frac{CP}{2}$$ ½

$$\therefore \ f = \frac{R}{2}$$

(a) With the help of a ray diagram, show how a concave mirror is used to obtain an erect and magnified image of an object.

(b) Using the above ray diagram, obtain the mirror formula and the expression for linear magnification.

CBSE 2018

(a) Calculate the distance of an object of height h from a concave mirror of radius of curvature 20 cm, so as to obtain a real image of magnification 2. Find the location of image also.

(b) Using mirror formula, explain why does a convex mirror always produce a virtual image.

CBSE 2016